

Hyperscale AI data centers

Hyperscale data centers are now powering AI models with a revolutionary architecture—at a staggering energy cost.

WHO: Amazon, Google, Meta, Microsoft, Nvidia, OpenAI

WHEN: Now

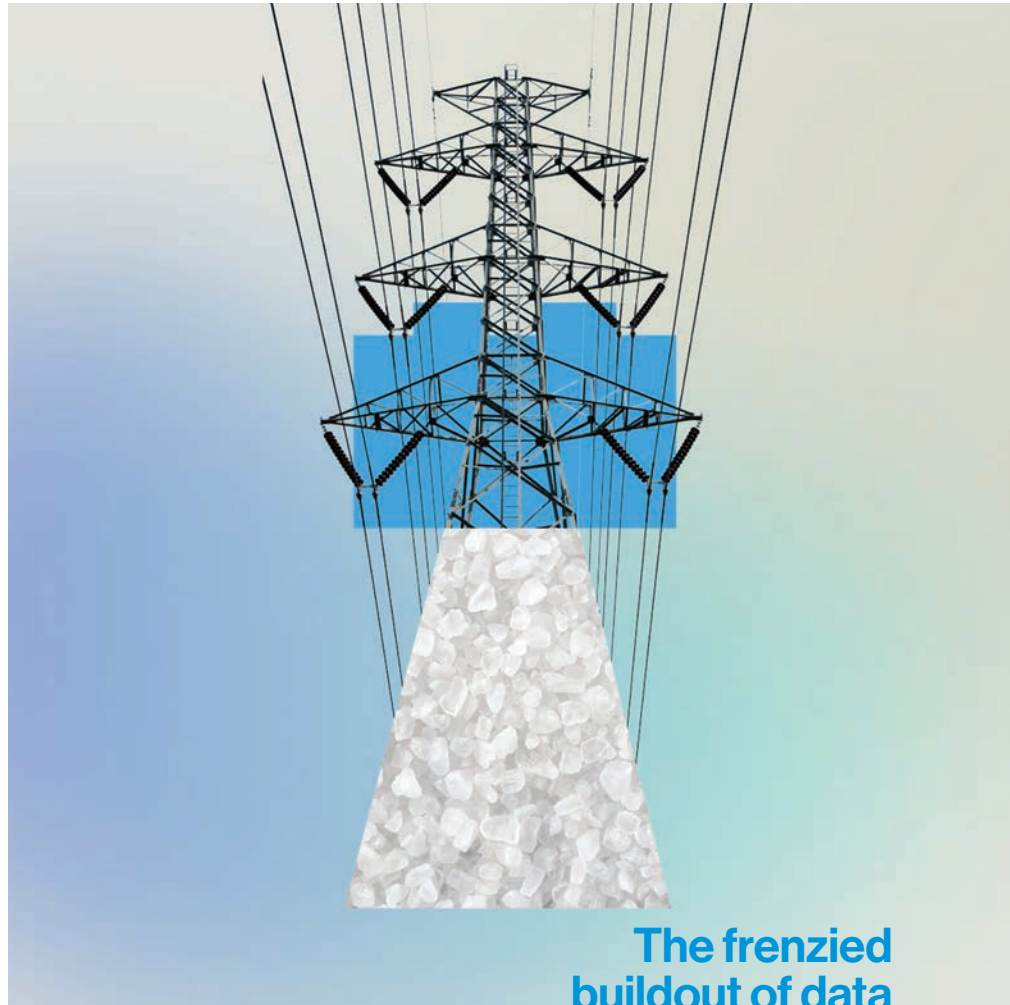
In sprawling stretches of farmland and industrial parks, super-sized buildings packed with racks of computers are springing up to fuel the AI race. These engineering marvels are a new species of infrastructure: supercomputers designed to train and run large language models at mind-bending scale, complete with their own specialized chips, cooling systems, and even energy supplies.

Hyperscale AI data centers bundle hundreds of thousands of specialized computer chips called graphics processing units (GPUs), such as Nvidia's H100s, into synchronized clusters that work like one giant supercomputer. These chips excel at processing massive amounts of data in parallel. Hundreds of thousands of miles of fiber-optic cables connect the chips like a nervous system, letting them communicate at lightning speed. Enormous storage systems continuously feed data to the chips as the facilities hum and whirl around the clock.

Tech companies like OpenAI, Google, Amazon, Microsoft, and Meta are pouring hundreds of billions of dollars into this infrastructure. Governments are spending big too.

But the impressive computing power comes at a cost. The densely packed chips run so hot that air-conditioning can't cool them. Instead, they're mounted to cold water plates or dunked in baths of cooling fluid. Dipping them in seawater may be next.

The largest data centers being built can devour more than a gigawatt of



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electricity—enough to power entire cities. Over half of that electricity comes from fossil fuels, while renewables meet just over a quarter of the demand. Some AI giants are turning to nuclear power. Google is dreaming of building solar-powered data centers in space.

The frenzied buildout of data centers is driven by the scaling laws of AI and by exploding demand as the technology gets wedged into everything from anime girlfriends to fitness apps. But the public may shoulder the costs of all this construction for years to come, as communities hosting the power-hungry facilities grapple with soaring energy bills, water shortages, droning noise, and air pollution.

—Michelle Kim